

User Manual

XRayVision AI — Getting Started and Complete Feature Guide

Field	Value
Document	User Manual
Version	1.0
Product version	2.1.0
Live application	https://x-ray-vision-board.vercel.app
Companion documents	SRS · SDD · Test Cases
Institution	Minhaj University Lahore — School of Software Engineering
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Supervisor	Maam Misbah — Lecturer, School of Software Engineering

Important Medical Disclaimer — read this first

XRayVision AI is an educational tool. It is NOT a certified medical device.

- Every result is an AI-generated estimate produced to **support**, never to replace, the clinical judgement of a qualified radiologist or physician.
- Do **not** use this software to make a diagnosis, to start or stop treatment, or to decide whether to seek care.
- The chatbot and the diet planner give general health information only. They are not a consultation.
- If you have a medical emergency, contact emergency services immediately — do not use this application.

An **EDUCATIONAL USE ONLY** badge is displayed permanently in the sidebar of every page, and a fuller disclaimer appears at the foot of every diagnostic report.

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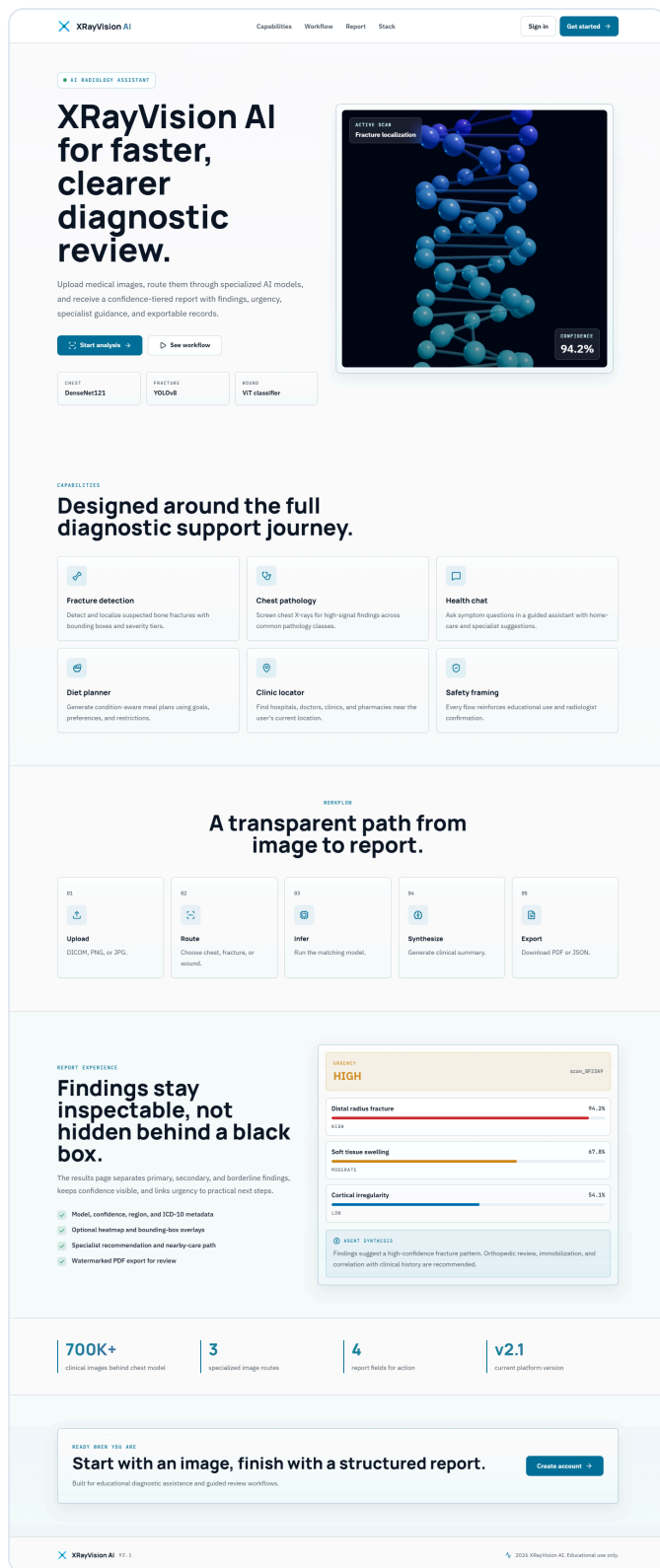
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1. What XRayVision AI Does

XRayVision AI takes a medical image you upload and returns a structured diagnostic report in seconds. Behind the scenes, four AI models work together — think of them as a team of four specialists:

Specialist	Model	What it does
 Chest specialist	DenseNet121	Screens chest X-rays for 18 pathologies, trained on 700,000+ clinical images
 Orthopaedic specialist	YOLOv8	Draws boxes around suspected fractures on bone X-rays
 Wound care specialist	ViT	Classifies photographs of external wounds
 Senior consultant	GLM 4.5 Air	Reads all their findings and writes one clear clinical summary with next steps

You choose which specialist to consult by selecting an **analysis route**. The system does not run every model on every image — sending a chest model a photograph of a wound would only produce noise.



The public landing page at <https://x-ray-vision-board.vercel.app>

2. Before You Begin

2.1 What you need

Requirement	Detail
Browser	Chrome, Edge, Firefox or Safari — a recent version
Internet	A stable connection. Images can be up to 20 MB.
An account	Free — see Section 3
Location permission	Only for the Clinic Locator. Optional.
Microphone permission	Only for chatbot voice input. Optional.

2.2 What images the system accepts

Property	Requirement
Formats	DICOM (.dcm), PNG, JPG
File size	Between 100 KB and 20 MB
Resolution	At least 200 × 200 pixels
Recommended	Clear, well-lit, non-blurry, at least 512 × 512 px for best accuracy

Why the minimum size? Very small or low-resolution images simply do not contain enough pixel detail for reliable inference. The system rejects them rather than returning an unreliable result.

3. Creating an Account and Signing In

3.1 Creating a new account

XRayVision AI

SECURE CLINICAL WORKSPACE

Diagnostic assistance with a clean chain of evidence.

Sign in to upload scans, review confidence-tiered findings, export reports, and continue care planning from one focused workspace.

- Multi-model analysis**
Chest, fracture, and wound routes stay separated for cleaner output.
- Educational guardrails**
Every report keeps clinical review and safety disclaimers visible.
- Private records**
Scan history and chat sessions are scoped to the authenticated user.

Educational use only. Not a substitute for a licensed radiologist.

ACCOUNT ACCESS

Create your account

Free for educational use — no credit card required

FULL NAME
Dr. Alex Reyes

EMAIL
you@hospital.org

PASSWORD
At least 8 characters

CONFIRM PASSWORD
Re-enter password

I AM A

☒ Medical Student ☐ Healthcare Professional ☐ Researcher ☐ Other

☐ I acknowledge that XRayVision AI is for educational and diagnostic assistance only, and is not a substitute for a licensed radiologist.

Create Account

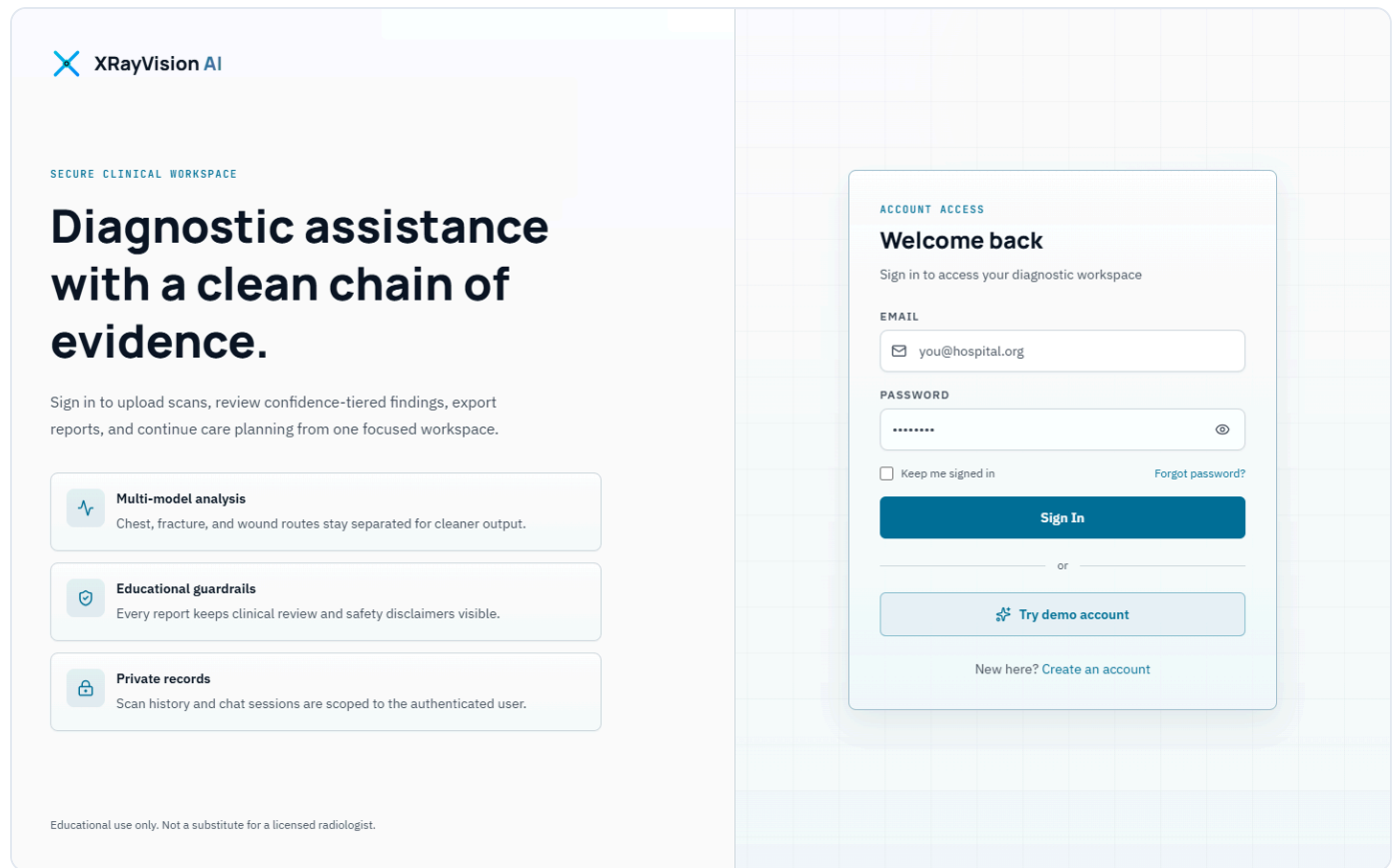
Already have an account? [Sign in](#)

1. Go to <https://x-ray-vision-board.vercel.app>.
2. Click **Get started** (top right) — or open `/auth/register` directly.
3. Fill in:
 - **Full name** — shown in the header and on your PDF reports
 - **Email** — your sign-in identifier
 - **Password** — must be **at least 8 characters**
 - **Confirm password** — must match exactly
4. Click **Create account**.

You are signed in automatically and taken straight to your Dashboard.

If you see "email already registered" — that address already has an account. Use [Sign in](#) instead, or [reset your password](#).

3.2 Signing in

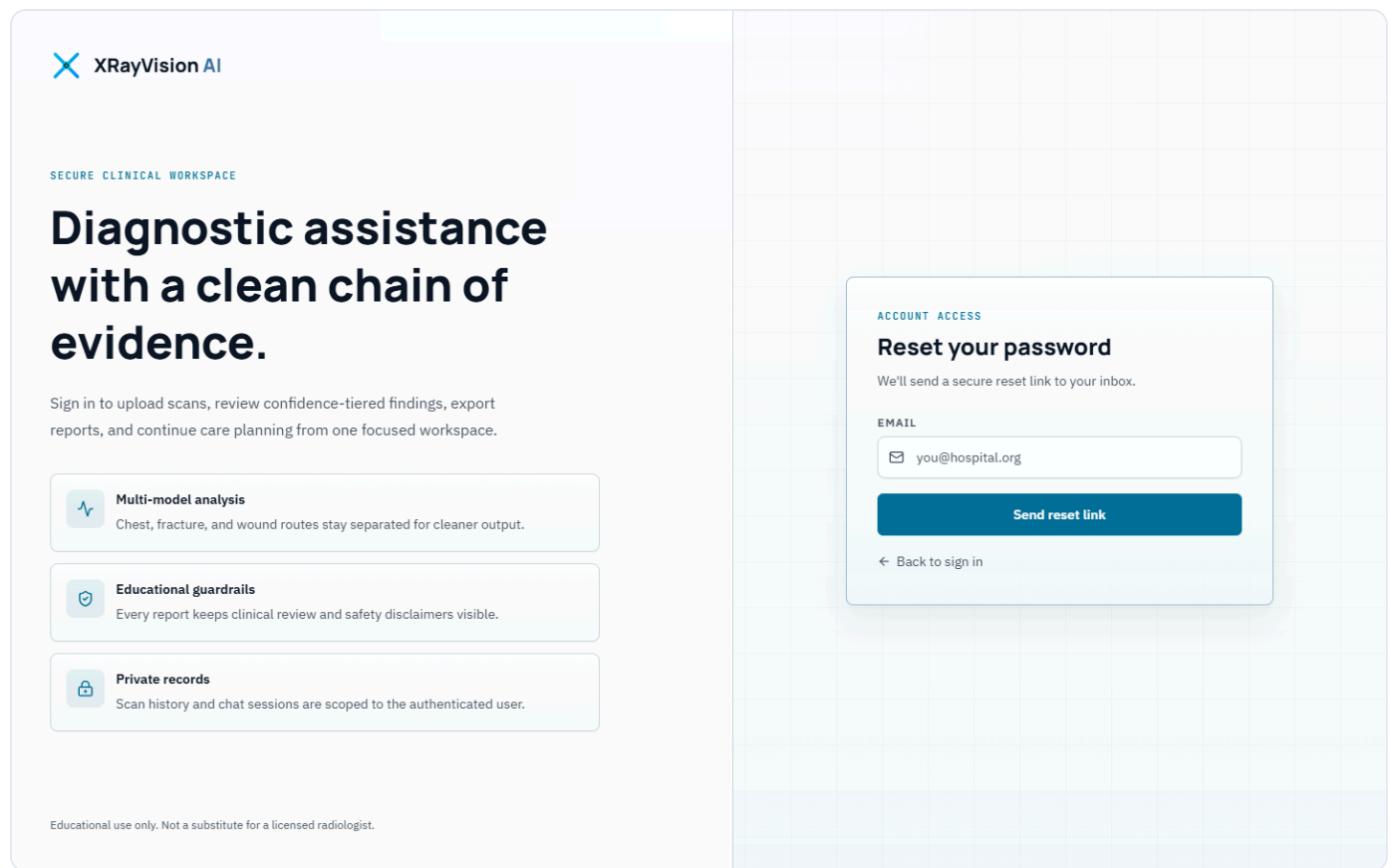


1. Open `/auth/login` or click **Sign in**.
2. Enter your email and password.
3. Click **Sign In**.

Trying the app without registering. Click **Try demo account** to sign in instantly to a pre-populated demonstration account. This is the fastest way for a reviewer or evaluator to explore every feature with real data already in place.

How long you stay signed in. Your session lasts **24 hours**. After that you will be returned to the sign-in page and will need to sign in again. This is a security measure.

3.3 Forgot your password?



1. On the sign-in page, click the **Forgot password** link.
2. Enter the email address of your account.
3. Click **Send reset link**.
4. Check your inbox — including the spam folder — and follow the link in the email.

4. Finding Your Way Around

Once signed in, every page shares the same layout.

The sidebar (left) — your main navigation:

Item	What it is for
Dashboard	Your statistics and recent activity
New Analysis	Upload an image and run the AI
Health Chat	Ask health questions in English or Urdu
Diet Planner	Generate a personalised 7-day meal plan
Clinics	Find hospitals and clinics near you

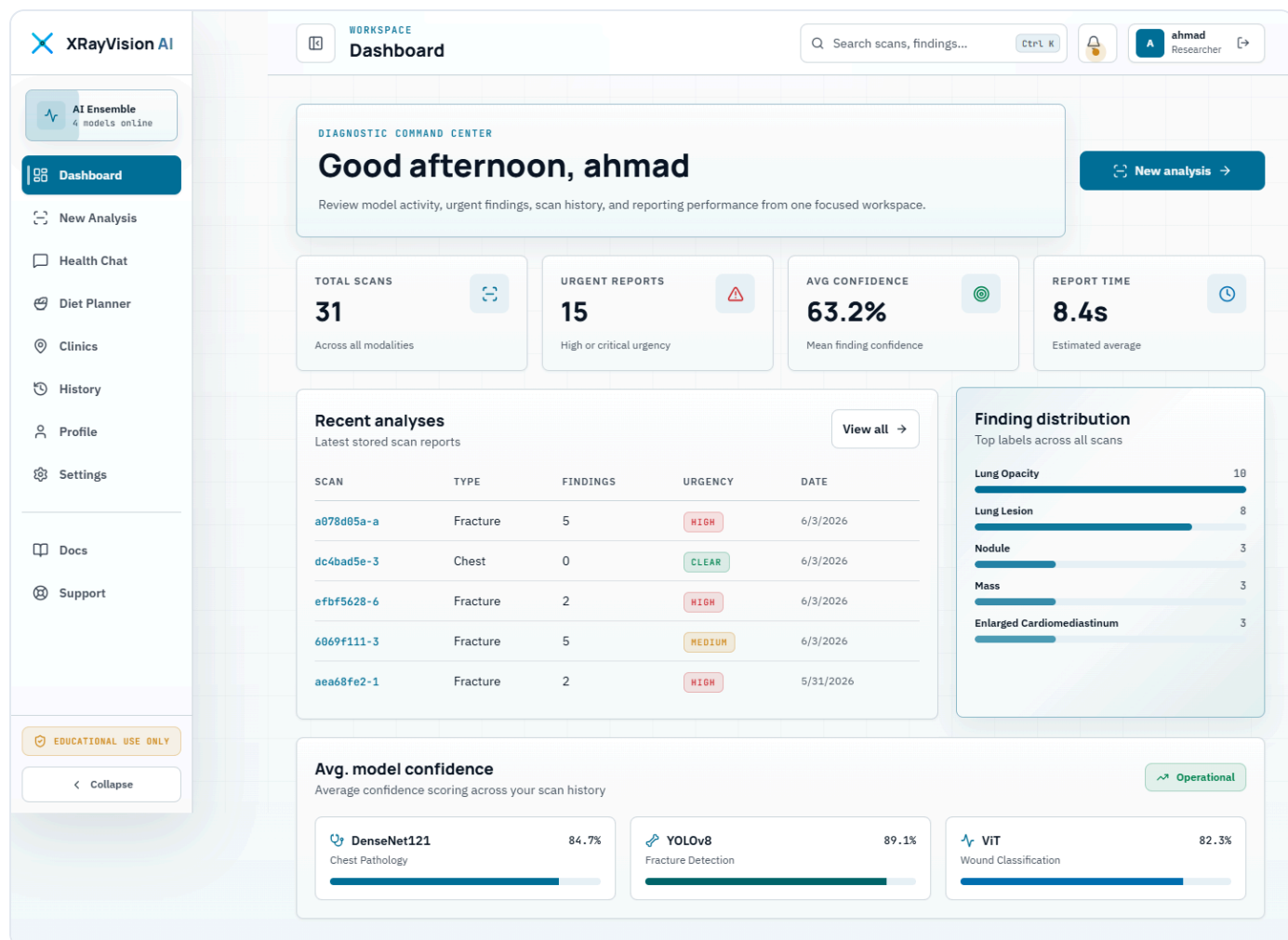
Item	What it is for
History	Every scan you have ever run
Profile	Your name, role and profile picture
Settings	Theme and display preferences

At the top of the sidebar, **AI Ensemble — 4 models online** shows the models are ready. At the bottom, the **EDUCATIONAL USE ONLY** badge is always visible. **Collapse** narrows the sidebar to give the content more room.

The header (top) — shows the current page name, a search box (**Ctrl K**), a notifications bell, and your account menu with the sign-out button.

5. The Dashboard

The Dashboard is your home page after signing in.



5.1 The four headline tiles

Tile	Meaning
Total scans	How many analyses you have run, across all image types
Urgent reports	How many of your scans came back rated high or critical
Avg confidence	The mean confidence score across all findings in your history
Report time	The average time your analyses took to complete

5.2 Recent analyses

The five most recent scans, with scan id, image type, number of findings, urgency badge and date. Click any row to open the full report. **View all** takes you to the complete History.

5.3 Finding distribution

A ranked bar chart of the labels that appear most often across your scans — useful for spotting patterns in the images you have been studying.

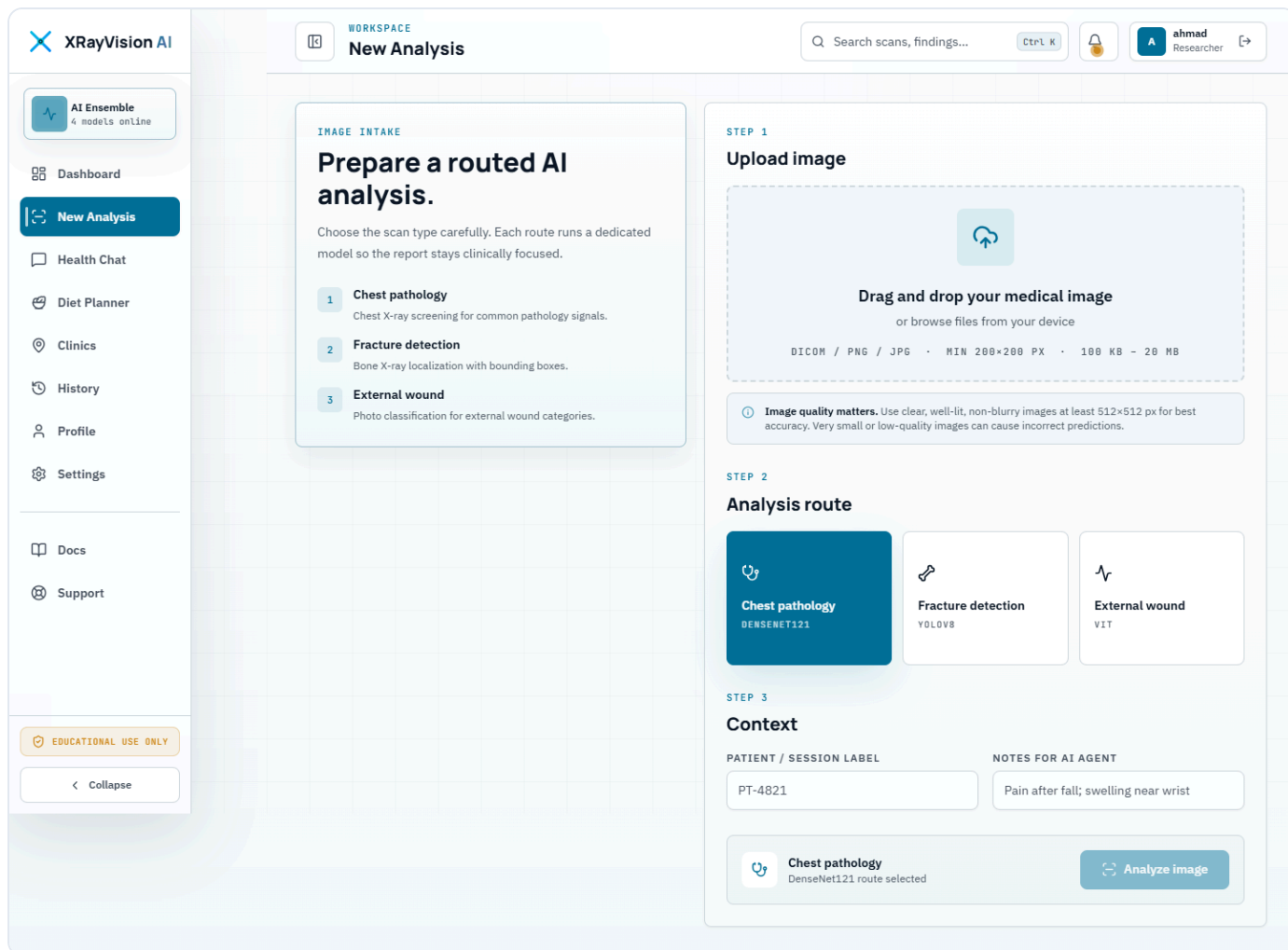
5.4 Average model confidence

How confident each of the three imaging models has been, on average, across your scan history.

These figures are yours alone. Every number on this page is calculated from only your own scans. No other user's data is ever included.

6. Running an Analysis (core feature)

This is the heart of the application. Click **New Analysis** in the sidebar.



The page guides you through **three steps**.

Step 1 — Upload your image

Drag your file onto the upload area, or click it to browse your device.

The panel restates the limits: DICOM / PNG / JPG • MIN 200×200 PX • 100 KB – 20 MB .

Image quality matters. Use clear, well-lit, non-blurry images of at least 512 × 512 px for best accuracy. Very small or low-quality images can cause incorrect predictions.

Step 2 — Choose your analysis route

Pick the one route that matches your image. This decides which AI model examines it.

Route	Model	Use it for
Chest pathology	DenseNet121	Chest X-rays — screens for common pathology classes
Fracture detection	YOLOv8	Bone X-rays — localises fractures with bounding boxes

Route	Model	Use it for
External wound	ViT	Photographs of external wounds

⚠️ **Choose carefully.** Selecting the wrong route gives a meaningless result. A chest model looking at a wound photograph has nothing useful to say about it.

Step 3 — Add context (optional but recommended)

Field	Purpose	Example
Patient / session label	A tag so you can find this scan later in History	PT-4821
Notes for AI agent	Clinical context passed to the AI, which uses it when writing the summary	Pain after fall; swelling near wrist

Notes genuinely change the output — the AI weighs your clinical context when assessing urgency and recommending next steps.

Running it

Click **Analyze image**. A full-screen processing overlay appears while the models work — this typically takes **8 to 15 seconds**.

First analysis of the day may be slower. The backend runs on a free-tier host that sleeps when idle. The very first request can take 30–60 seconds while the AI models load into memory. Subsequent analyses are fast.

When it finishes you are taken automatically to your report.

7. Reading Your Diagnostic Report

The report has two columns.

7.1 Left column — the image

Your uploaded image is displayed with the AI's findings drawn on top as dashed boxes, each labelled with its finding name and confidence.

The controls beneath the image:

Control	Effect
AI Findings	Show or hide the bounding boxes
Heatmap	Show or hide the confidence heatmap overlay
Labels	Show or hide the text labels on each box
 + / -	Zoom between 0.5× and 2.5×
	Reset the zoom to normal

7.2 Right column — the findings

The **urgency banner** at the top gives the overall assessment, in one of five levels:

Level	Meaning
CRITICAL	Findings suggest an immediately serious condition
HIGH	Significant findings requiring prompt professional attention
MEDIUM	Findings warranting follow-up
LOW	Minor or uncertain findings
CLEAR	No significant findings detected

Each level is shown as **both** a colour and a word, so the meaning is never ambiguous.

The low-confidence warning. If any finding scored below 60 % confidence, a blue banner appears: *"One or more findings have confidence below 60% — radiologist review recommended."* Take this seriously — it means the AI itself is uncertain.

Findings, grouped by confidence. This grouping is deliberate — it makes uncertainty visible instead of leaving you to interpret raw percentages:

Group	Confidence	How it is shown
Primary Findings	65 % and above	Full cards with confidence bars — the AI's strongest signals
Secondary Findings	50 % – 65 %	Compact rows, labelled with the confidence range
Borderline	Below 50 %	Collapsed behind a "click to expand" control — weak signals, shown for completeness

Each finding card shows the finding name, confidence percentage, severity, the model that produced it, the anatomical region, and an ICD-10 diagnostic code where one is mapped.

GLM Agent Analysis. The AI's written clinical summary — what the findings suggest, taken together, and what the confidence levels mean in context. It always ends by reminding you that this is educational, not a substitute for a qualified radiologist.

Immediate Actions. A numbered list of practical next steps — consultations, further imaging, management measures, follow-up.

Recommended specialist. Which kind of doctor these findings would normally be referred to, with a **Find nearby** → link that takes you straight to the Clinic Locator.

Confidence Summary. A complete table of every finding — model, finding, confidence, severity — so nothing is hidden behind the tiered grouping.

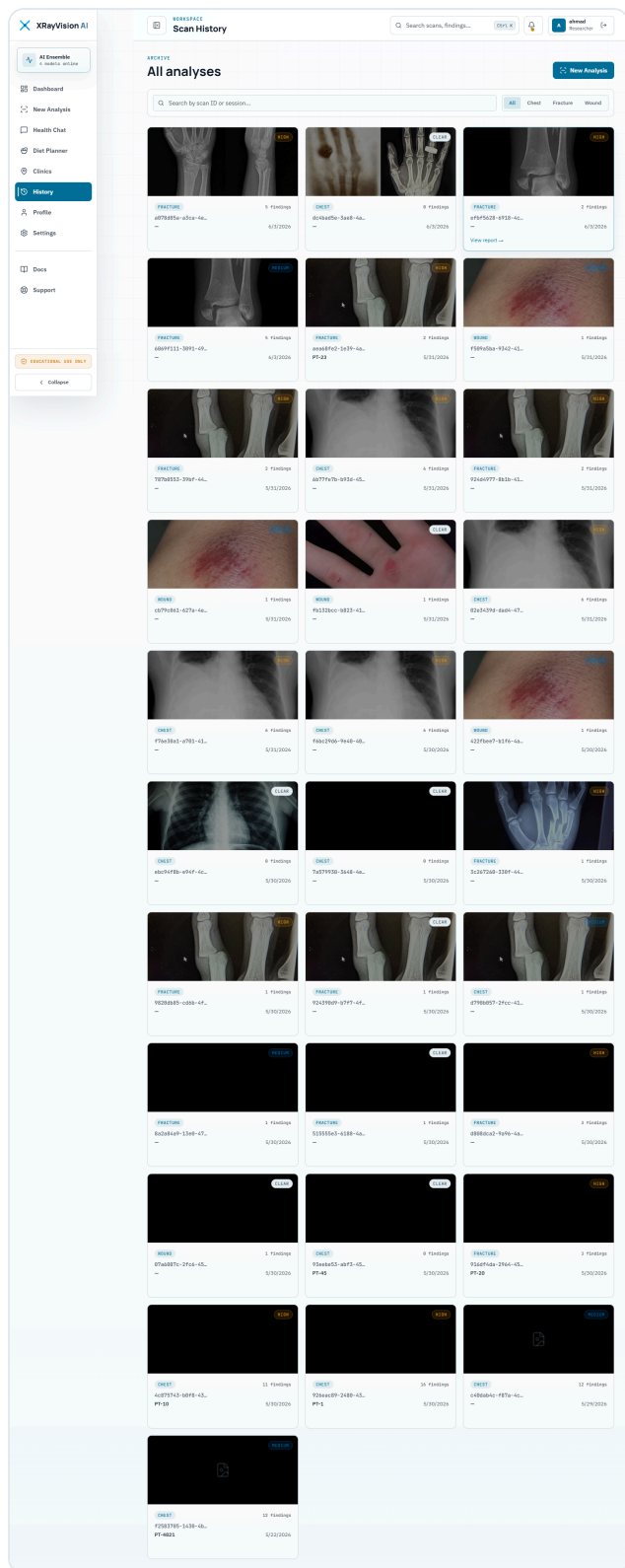
7.3 Saving your report

At the bottom of the report:

- **Download PDF Report** — a formatted document with the image, findings, summary and disclaimer. This is the version to attach to a referral or a case study.
 - **Export JSON** — the raw structured findings, for further analysis or research.
 - **Back to history** — returns to your scan list.
-

8. Scan History and Exports

Click **History** in the sidebar to see every analysis you have ever run.



Scans are listed **newest first**. Each row shows the image thumbnail, scan type, number of findings, urgency badge and date. Hovering a row reveals a **View report** → link.

What you can do here:

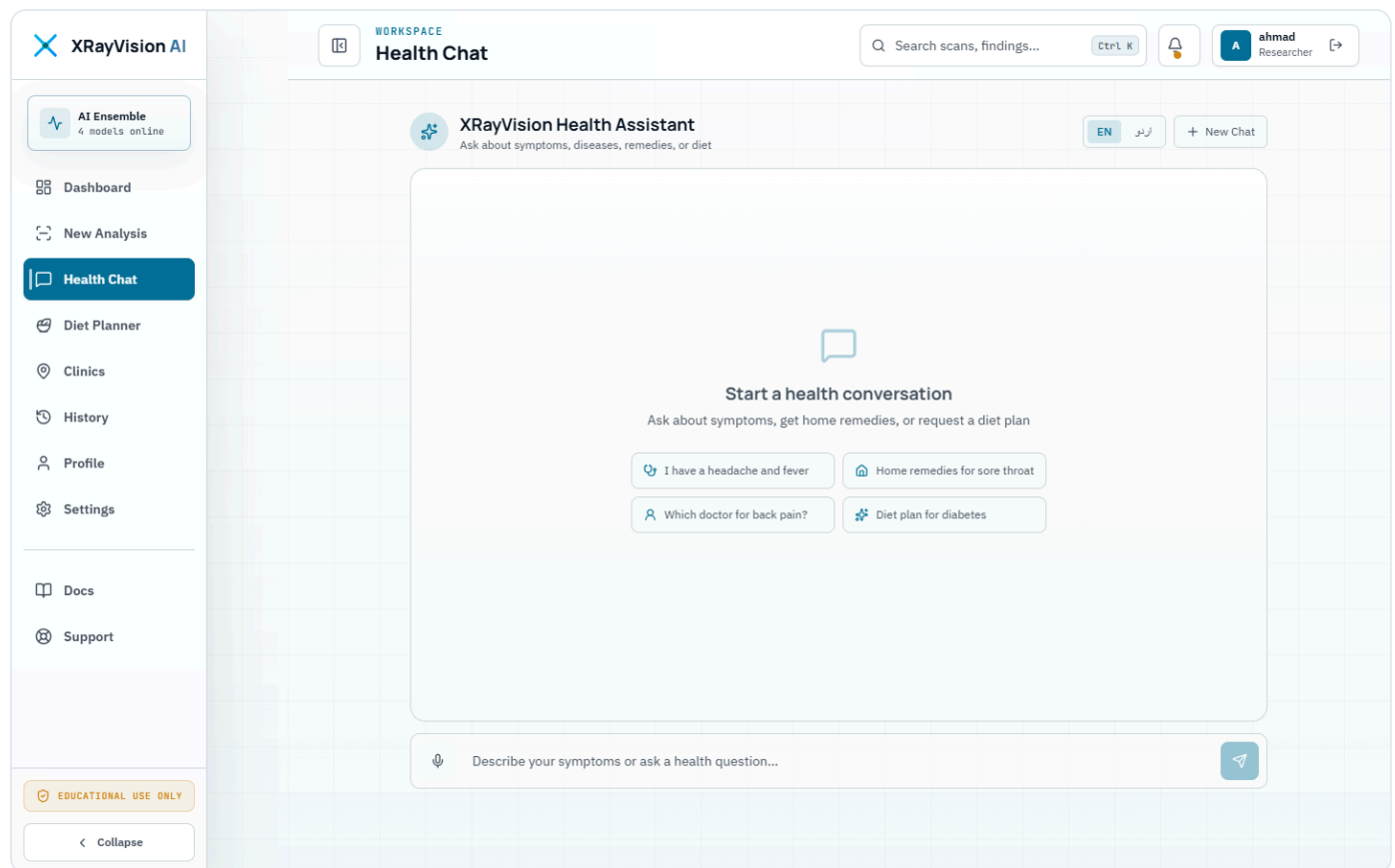
Action	How
Open a report	Click the row
Filter by type	Use the scan-type filter to show only chest, fracture or wound scans

Action	How
Download a PDF	Open the report, then click Download PDF Report
Export JSON	Open the report, then click Export JSON
Delete a scan	Use the delete control on the scan

⚠ **Deletion is permanent.** Deleting a scan removes both the database record and the stored image. There is no undo and no recycle bin. Download the PDF first if you might need it.

9. The Health Chatbot

Click **Health Chat** for general medical questions in **English or Urdu**.



9.1 Asking a question

1. Choose your language with the **EN / اردو** toggle at the top.
2. Type your question in the box at the bottom — or click the microphone to dictate it.
3. Press Enter or click send.

When you select Urdu, the interface switches to right-to-left and the placeholder changes to اپنی علامات بتائیں..

9.2 What you get back

Alongside the written reply, the assistant surfaces two structured cards where relevant:

- **Doctor type** — the kind of specialist your described symptoms would normally be taken to
- **Home remedies** — general self-care suggestions

9.3 Conversation memory

The chatbot remembers the last **10 messages** in your conversation, so you can ask follow-up questions naturally:

You: I've had a fever for two days. **Bot:** *(replies about fever)* **You:** How long should it last? ← the bot knows you still mean the fever

Your conversations are saved. When you come back to the Health Chat page, previous sessions are listed and can be reopened.

 **What the chatbot will not do.** It will never give you a definitive diagnosis, and it will always point you toward a qualified professional for real medical decisions. This is deliberate. If you press it for a yes-or-no answer about a serious condition, it will decline and refer you to a doctor.

10. The Diet Planner

Click **Diet Planner** to generate a personalised **7-day meal plan**.

10.1 Filling in the form

Field	What to enter	Example
Medical condition	Any condition the plan should account for	diabetes , high blood pressure
Dietary preference	Your general eating style	balanced , vegetarian
Restrictions	Allergies and foods to avoid	gluten-free, dairy-free, nut allergy
Goals	What you want the plan to achieve	weight loss , muscle gain

Click generate and the plan appears below.

10.2 Built-in medical safety rules

The planner applies condition-specific clinical guardrails automatically:

If you enter	The plan applies
Hypertension / high blood pressure	DASH dietary principles — reduced sodium, more fruit, vegetables and low-fat dairy
Diabetes	Low-glycaemic-index food choices
Kidney disease	An explicit recommendation to consult a renal dietitian — kidney nutrition needs individual professional supervision

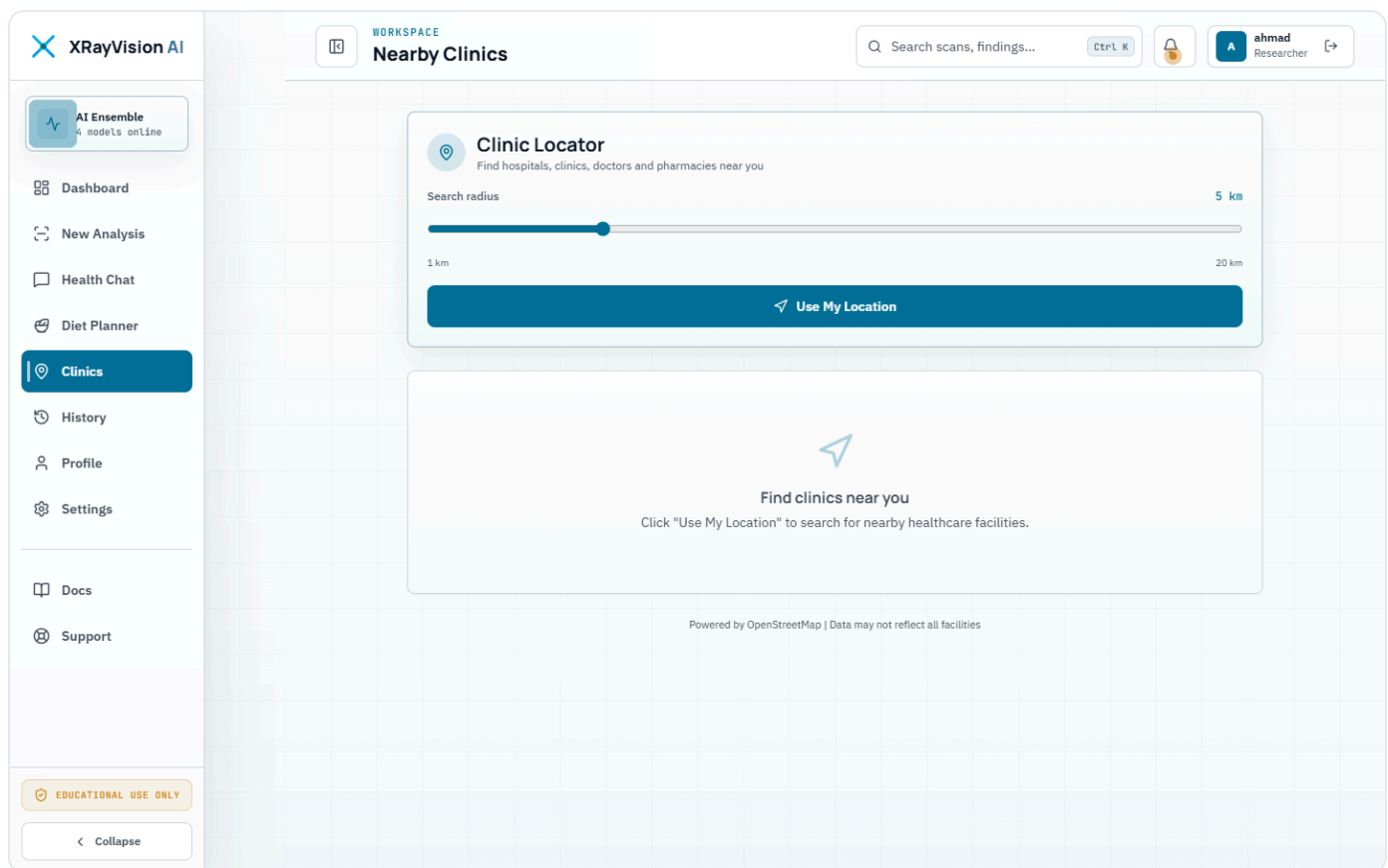
10.3 Your plan

Each of the seven days shows **Breakfast**, **Lunch** and **Dinner** — with a name, a description and, where available, calories and key nutrients. Snacks are included where appropriate, along with a summary and general tips.

⚠ General guidance only. If you have a medical condition, discuss any diet plan with your doctor or a qualified dietitian before following it.

11. Finding Nearby Clinics

Click **Clinics** to find hospitals, clinics, doctors and pharmacies near you.



1. Set your **search radius** with the slider — anywhere from **1 km to 20 km**. It starts at 5 km.
2. Click **Use My Location**.
3. Your browser will ask permission to share your location. Click **Allow**.

Results appear sorted by distance, nearest first. Each shows the facility name, its type, its address, how far away it is, and a link that opens it in Google Maps.

Facility data comes from **OpenStreetMap**, a community-maintained map database. The page notes: "*Data may not reflect all facilities.*"

If you decline location permission, the page explains what happened and no search is run. You can grant permission in your browser's site settings and try again.

If no results appear, increase the radius with the slider and search again. Coverage varies by region.

12. Your Profile

Click **Profile** to view and edit your account details.

The screenshot displays the XRayVision AI Profile page. The left sidebar contains navigation links: Dashboard, New Analysis, Health Chat, Diet Planner, Clinics, History, Profile (selected), Settings, Docs, and Support. The main content area is titled 'Profile' and shows account details for 'ahmad' (ahmad123@gmail.com, Member since 5/22/2026). The profile card includes a role of 'Researcher', 31 scans, and an average confidence of 63.2%. Below this, the 'Personal details' section allows editing of full name, email, and role, with a 'Save Changes' button. The 'Usage Statistics' section shows 31 total scans, 15 critical findings, and 63.2% average confidence. A 'Session' section at the bottom provides a 'Sign out' button.

Here you can:

- Change your **display name** — this appears in the header and on your PDF reports
- Update your **role** (for example Medical Student, Radiologist, Researcher) — this is a descriptive label only and does not change what you can access
- Upload a **profile picture**

Your scan statistics are also summarised on this page.

13. Settings

Click **Settings** to personalise how the application looks and behaves.

XRayVision AI

Settings

Search scans, findings... Ctrl K

ahmad Researcher

Workspace Controls

Personalize your diagnostic workspace

Theme, analysis defaults, overlays, language, and notifications are stored with your profile.

Appearance

Choose the visual mode and interface language.

THEME

Light Dark System ✓

LANGUAGE

English Urdu

Analysis Preferences

Set defaults for model confidence and visual overlays.

CONFIDENCE THRESHOLD

40%

More sensitive More conservative

Show AI detection boxes on images

Show confidence heatmap overlay

Notifications

Control alert behavior for urgent reports.

Email notifications for critical findings

Settings apply immediately in this browser and persist after saving.

Save Settings

13.1 Appearance

Choose **Light**, **Dark** or **System** (which follows your operating system's setting).

13.2 Analysis preferences

Setting	Effect
Show AI detection boxes on images	Whether bounding boxes are drawn by default on result images
Show confidence heatmap overlay	Whether the heatmap layer is shown by default

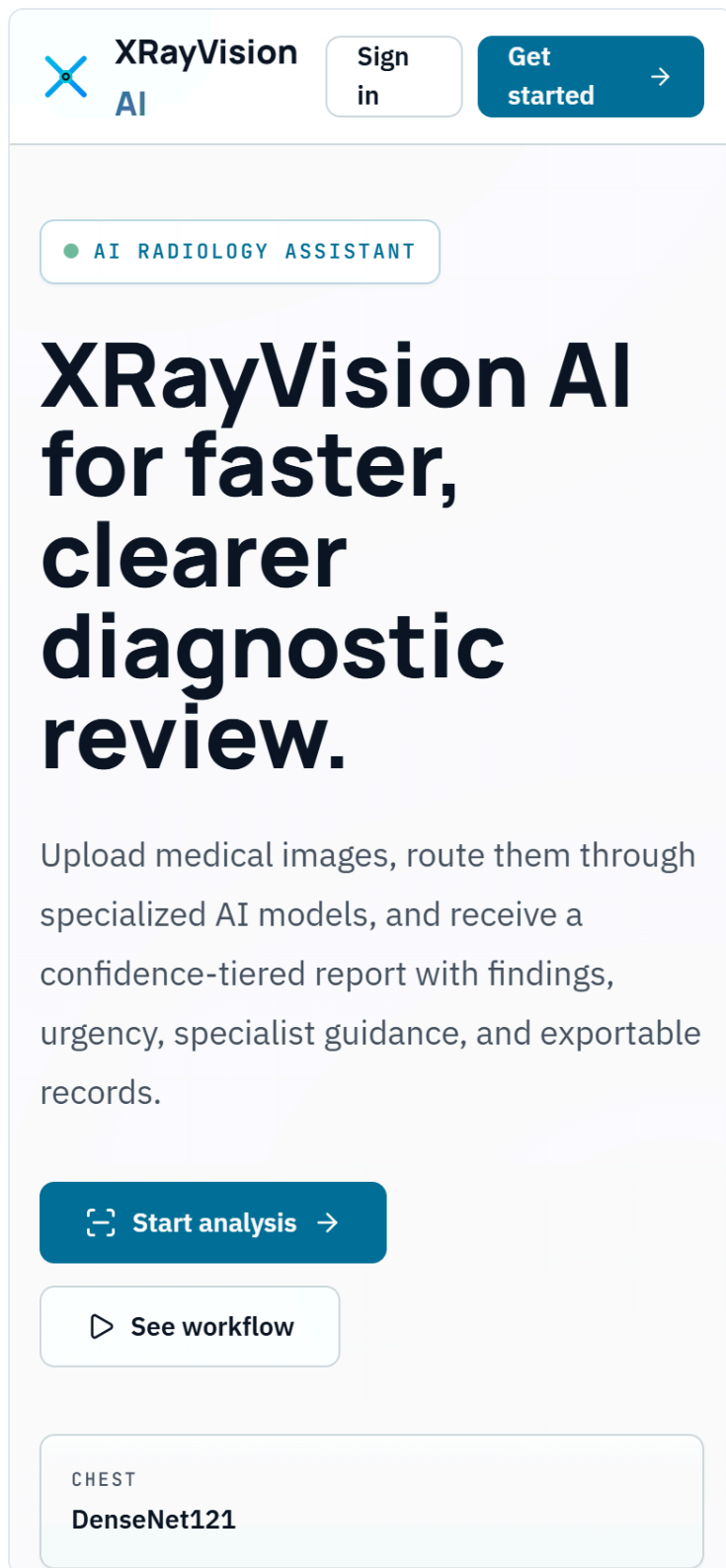
13.3 Notifications

Setting	Effect
Email notifications for critical findings	Whether you are emailed when a scan returns a critical result

Your preferences are saved to your account, so they follow you to any device you sign in from.

14. Using XRayVision AI on a Phone

The application is fully responsive and works on mobile browsers.



On a small screen the layout reflows to a single column and the sidebar becomes a menu. Everything works, though reviewing bounding boxes on a detailed X-ray is naturally easier on a larger screen.

15. Troubleshooting

Upload and analysis

Message or symptom	What it means	What to do
"File is too small (40 KB). Minimum is 100 KB..."	The image lacks enough pixel data for reliable inference	Use a higher-quality export of the image
"File is too large (24.3 MB). Maximum allowed size is 20 MB."	The file exceeds the upload limit	Compress or resize the image below 20 MB
"Image resolution is too low (150×150 px)..."	Below the 200 × 200 px minimum	Use an image of at least 200 × 200 px — 512 × 512 px or larger is recommended
"File could not be decoded as a valid image..."	The file is not a readable JPEG, PNG or DICOM	Check the file actually opens in an image viewer; renaming a PDF to .jpg will not work
"Empty file uploaded."	The file contains no data	Re-export or re-download the image
"Rate limit exceeded. Please slow down."	More than 10 analyses in one minute	Wait a minute and try again
Analysis failed screen	The AI models could not process the image	Verify you selected the right route for the image type, then retry. If it persists, the backend may be restarting — wait a minute.
"AI synthesis temporarily unavailable. Please review findings manually."	The AI models worked, but the summary-writing service was unreachable	The findings are still valid — read them directly. Re-run later for a written summary.
The first analysis takes 30–60 seconds	The free-tier backend was asleep and is loading models	Normal. Subsequent analyses are fast.
Results appear but the image is missing	Image storage was unavailable during upload	The findings are still valid. Re-run the analysis if you need the image saved.

Signing in

Symptom	What to do
Signed out unexpectedly	Sessions last 24 hours. Simply sign in again.
"Invalid email or password"	Check for typos and caps lock. If you still cannot get in, use Forgot password .
"Demo login failed. Please try again."	The backend may be waking up. Wait 30 seconds and retry.
Reset email never arrived	Check your spam folder, and confirm you entered the address the account was registered with.

Other features

Symptom	What to do
Clinic search finds nothing	Increase the radius. OpenStreetMap coverage varies by region.
Clinic search shows an error	The public map service may be busy. Wait a moment and try again.
Location permission blocked	Grant location access for this site in your browser's site settings.
Microphone does not work in chat	Grant microphone permission, or just type your question instead.
Chatbot replies are slow	Replies typically take a few seconds. Longer waits usually mean the AI service is busy.
Diet plan looks generic	If the AI's response failed validation, a safe pre-written plan is shown instead. Try again with more specific inputs.

16. Frequently Asked Questions

Can I use this to diagnose a patient? No. XRayVision AI is an educational tool and is not a certified medical device. Every output is an AI estimate to support learning and preliminary review. Clinical decisions must be made by a qualified professional.

How accurate is it? Accuracy varies by model, by image quality and by the condition being examined. That is precisely why every finding shows a confidence percentage, why findings are grouped into confidence tiers, and why anything below 60 % triggers an explicit recommendation for radiologist review. Treat low-confidence findings as prompts to look more carefully — not as conclusions.

What happens to the images I upload? They are stored in your account so you can review them later, and are only ever shown alongside your own scans. Deleting a scan removes both the record and the image permanently.

Can other users see my scans? No. Every scan, chat conversation and statistic is tied to your account. Ownership is checked in the application and again in the database.

Why do I have to sign in again every day? Sessions expire after 24 hours as a security measure.

Why was my image rejected? Almost always size or resolution — the minimums are 100 KB and 200 × 200 px. The error message always states the specific reason and the limit involved.

Can I upload a DICOM file straight from a scanner? Yes. DICOM (.dcm) files are supported alongside PNG and JPG.

Which route should I pick? Chest X-ray → **Chest pathology**. Bone X-ray where you suspect a fracture → **Fracture detection**. Photograph of a wound on the skin → **External wound**.

Why did I get no findings at all? No finding met the confidence threshold. The urgency will read **CLEAR**. This does not prove the image is normal — it means this model found nothing it was confident about.

Is the chatbot a doctor? No. It provides general health information and points you toward the right kind of specialist. It will not diagnose you.

Is it free? Yes. The entire platform runs on free-tier services. That is also why the first request after a period of inactivity can be slow.

17. Glossary

Term	Plain-English meaning
Bounding box	The rectangle the AI draws around a region it thinks is abnormal
CLAHE	A contrast-enhancement step applied to your image before analysis, so faint details become easier for the model to see
Confidence	How sure the model is, as a percentage. Higher is more certain — but it is never a guarantee.
DenseNet121	The neural network that examines chest X-rays, trained on over 700,000 clinical images
DICOM	The standard file format that medical imaging equipment produces (<code>.dcm</code>)
Finding	One thing the AI thinks it has spotted, with a name and a confidence score
ICD-10 code	The internationally standard code for a diagnosis, e.g. <code>S02-S92</code> for certain fractures
LLM	Large Language Model — the AI that writes the readable clinical summary
Severity	How serious a single finding is judged to be: critical, high, moderate or low
Synthesis	The AI-written paragraph that pulls all the findings together into one clinical picture
Urgency	The overall rating for the whole scan: critical, high, medium, low or clear
ViT	Vision Transformer — the neural network that classifies wound photographs
YOLOv8	The object-detection model that finds and boxes fractures on bone X-rays

Getting Help

Need	Where to go
Technical API reference	<code>/docs</code> on the backend (Swagger UI)

Need	Where to go
What the system is required to do	Software Requirements Specification
How the system is built	Software Design Document
How it was verified	Test Documentation
Reporting a security issue	SECURITY.md
Contributing	CONTRIBUTING.md

Revision History

Version	Date	Author	Description
1.0	August 2026	Muhammad Ali Raza, Hamza Afzal	Initial user manual with screenshots from the live deployment

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